

A Theoretical Analysis

We now show that SPLICE terminates in a finite number of profit evaluations and returns a solution with a formally characterized optimality property.

A.1 SPLICE Convergence

We begin with defining κ -local optimum and use it to establish convergence conditions.

Definition A.1 (κ -local optimum). A subset $S \subseteq \mathcal{S}$ is a κ -local optimum of $\mathcal{P}_{\mathcal{T}}$ if no swap of up to κ sources improves profit, i.e. for all $R \subseteq S$ and $D \subseteq \mathcal{S} \setminus S$ with $|R| = |D| \leq \kappa$:

$$\mathcal{P}_{\mathcal{T}}((S \setminus R) \cup D) \leq \mathcal{P}_{\mathcal{T}}(S). \quad (4)$$

THEOREM A.2 (CONVERGENCE). Let $\mathcal{P}_{\min}$ and $\mathcal{P}_{\max}$ be finite lower and upper bounds on $\mathcal{P}_{\mathcal{T}}$ over $2^{\mathcal{S}}$. Introduce a minimum profit improvement $\varepsilon > 0$ required to accept a swap. For fixed $k_{\max}$ and $s_{\max}$, SPLICE terminates after at most

$$O\left(s_{\max} \cdot k_{\max} \cdot n \cdot \frac{\mathcal{P}_{\max} - \mathcal{P}_{\min}}{\varepsilon}\right)$$

profit evaluations. Furthermore, if SPLICING exhaustively evaluates all k -swaps (rather than the marginal-gain heuristic), the returned subset is a $k_{\max}$ -local optimum of $\mathcal{P}_{\mathcal{T}}$ (Definition A.1).

In our implementation, every strict improvement is accepted; ε serves as an analytical parameter abstracting the minimum representable positive profit increment on a finite dataset.

PROOF. SPLICING accepts a candidate S' only when $\mathcal{P}_{\mathcal{T}}(S') \geq \mathcal{P}_{\mathcal{T}}(S_{\text{curr}}) + \varepsilon$, so profit increases by at least ε at each accepted step. Since $\mathcal{P}_{\mathcal{T}}$ is bounded in $[\mathcal{P}_{\min}, \mathcal{P}_{\max}]$, the number of accepted steps per FIXEDSUPPORT run is at most $(\mathcal{P}_{\max} - \mathcal{P}_{\min})/\varepsilon$. At each step, SPLICING iterates over $k = 1, \dots, k_{\max}$. For each k , it computes marginal gains $\Delta^-(s)$ for every $s \in S$ ($|S| \leq n$ evaluations) and $\Delta^+(u)$ for every $u \in \mathcal{S} \setminus S$ ($n - |S|$ evaluations), then evaluates the single proposed swap (1 evaluation): $O(n)$ evaluations per k , and $O(k_{\max} \cdot n)$ per SPLICING call. Over all accepted steps and $s_{\max}$ initializations, total evaluations are bounded by $O(s_{\max} \cdot k_{\max} \cdot n \cdot (\mathcal{P}_{\max} - \mathcal{P}_{\min})/\varepsilon)$. SPLICE terminates when no proposed k -swap improves profit by ε for any $k \leq k_{\max}$, i.e. $S_{\text{curr}} = S_{\text{prev}}$ in FIXEDSUPPORT. If SPLICING exhaustively evaluates all $\binom{|S|}{k} \binom{n-|S|}{k}$ candidate k -swaps rather than using the marginal-gain heuristic, this termination condition is equivalent to Definition A.1, and the returned set is a $k_{\max}$ -local optimum. The practical algorithm uses marginal gains to guide swap selection, trading the formal local-optimum guarantee for the $O(k_{\max} \cdot n)$ per-step cost. $\square$

Practical implication. We observe in experiments (Section 5) that SPLICE converges in 307–6,951 evaluations (Figure 3), far below the exhaustive worst case of $O(n^{2k_{\max}})$, because the profit landscape has few local optima at the operating $k_{\max} = 5$ and generating the active set with the top- i sources ranked by individual profit already starts the search in high-profit regions of the subset space.

A.2 Submodularity Ratio of the Profit Landscape

Theorem A.2 guarantees termination at a local optimum but says nothing about its distance from the global optimum. We characterize the structure of $\mathcal{P}_{\mathcal{T}}$ via the *submodularity ratio* of Das and Kempe [9], used here as a diagnostic.

Definition A.3 (Submodularity ratio [9]). The submodularity ratio of $\mathcal{P}_{\mathcal{T}}$ with respect to ground set U (the universe of items over which $\mathcal{P}_{\mathcal{T}}$ is defined) and parameter κ is

$$\gamma_{U,\kappa} = \min_{\substack{L \subseteq U \\ S \subseteq U \setminus L \\ |S| \leq \kappa}} \frac{\sum_{i \in S} [\mathcal{P}_{\mathcal{T}}(L \cup \{i\}) - \mathcal{P}_{\mathcal{T}}(L)]}{\mathcal{P}_{\mathcal{T}}(L \cup S) - \mathcal{P}_{\mathcal{T}}(L)}.$$

In our setting, $U = \mathcal{S}$, the full set of candidate sources. Das and Kempe's original definition assumes monotone $\mathcal{P}$, under which $\gamma \in [0, 1]$ ($\gamma = 1$ for submodular functions), and show that a greedy solution achieves a $(1 - e^{-\gamma})$ approximation for monotone objectives. Our profit function is non monotone because adding a source is not guaranteed to increase utility; therefore, the formal Das-Kempe interpretation does not apply and the approximation guarantee does not transfer to our setting. We, therefore, report the ratio as a qualitative landscape diagnostic rather than a formal bound, with values > 1 indicating that sum of the individual gains exceeds the combined gain, and values < 0 indicating that the individual and combined gains disagree. We report the results from our empirical estimation in Section A.3.

A.3 Empirical estimation of submodularity ratio

As discussed in Section A.2, for each benchmark, we estimate $\gamma_{\mathcal{S},5}$ – i.e., the submodularity ratio with ground set equal to the full candidate pool $\mathcal{S}$ and parameter $\kappa = 5$ – by sampling 500 random subset pairs (L, S) with $|S| = 5$. For each pair, we compute the ratio of the sum of individual marginal gains to the combined gain. Because the joint gain can be zero or negative in our non-monotone setting (in which case the ratio is undefined or numerically unstable), we filter such pairs and report the median and the fraction of remaining pairs falling into the regimes stated in Section A.2.

Table 7 reports the results. All benchmarks have a median ratio above 1, indicating that substitution dominates the typical landscape: in 53–78% of sampled pairs, the sum of individual contributions exceeds the joint contribution, reflecting overlapping signal across sources. Two patterns vary meaningfully across benchmarks. First, the median itself varies by a factor of 2.3 $\times$, from a barely substitution flavored ACSTravelTime (median 1.07) to the strongly substitution flavored ACSPublicCoverage (2.37) and IPO (2.43). Second, the opposite sign fraction ranges from 8% on Amazon to 33% on ACSTravelTime, characterizing how often individual sources have effects of opposite sign to the joint effect. The fraction of sampled pairs with non-positive joint gain – those filtered before ratio computation – ranges from 12% on Wilds (where source diversity makes most multi source additions broadly useful) to 47% on ACSPubCoverage (where source cancellation is most prevalent).

These diagnostic patterns track SPLICE's empirical advantage over greedy forward selection. On benchmarks with high median and low opposite sign fraction (ACSPubCoverage, IPO, Amazon), the landscape rewards local greedy choices, and greedy reaches profit within 0.5–2.2 points of SPLICE. On ACSTravelTime, the lowest median benchmark with the highest opposite sign fraction,

Table 7: Submodularity ratio diagnostics across benchmarks. *Substitutes* ($\gamma > 1$, sum of individual gains exceeds joint gain), *Partial-complements* ($0 < \gamma \leq 1$, joint gain exceeds or matches the sum), *Opposite-sign* ($\gamma < 0$, individual gains and joint gain have opposite signs, only possible in non-monotone settings). % *Filtered* reports the fraction of all 500 sampled pairs filtered before γ computation because the joint gain was non-positive.

Benchmark	Median γ	% Sub	% Part-comp	% Opp-sign	% Filtered
ACSIncome	1.97	77	11	11	26
ACSPubCov	2.37	76	11	11	47
ACSTravelTime	1.07	53	14	33	41
Amazon	1.71	78	14	8	24
IPO	2.43	69	10	21	39
Wilds	1.44	69	14	17	12

greedy gets stuck in local optima that 1-step additions cannot escape — SPLICE exceeds greedy by 18 MSE points. On Wilds, the substitution structure is moderate but the per-source value variance is large (single-source macro-F1 spans 0.05–0.20), so multistart initialization and k-swap together are necessary to navigate the landscape — SPLICE outperforms greedy by 5.8 macro-F1 points, the largest gap in our suite. The submodularity ratio diagnostic is consistent with where SPLICE’s mechanism is most beneficial, even though it does not transfer formal approximation guarantees to the non-monotone setting.

B Reference Models for SPLICE-GRAD

SPLICE-GRAD uses a task-specific reference model trained on the union of all source data before gradient extraction. Architectures are selected to be lightweight (so gradient computation is fast) while remaining expressive enough that gradient directions are informative.

Tabular classification (ACSIncome, ACSPubCoverage, SANTOS). An ℓ_2 -regularized logistic regression model (`nn.Linear( $p$ , 1)` with sigmoid output), trained with binary cross-entropy for 5 epochs. Gradient dimension: $d_g = p + 1$, where p is the number of input features.

Tabular regression (ACSTravelTime). A single linear layer (`nn.Linear( $p$ , 1)`), trained with mean squared error for 5 epochs. Gradient dimension: $d_g = p + 1$.

Image classification (Wilds). A linear head `nn.Linear( $d_{\text{feat}}$ ,  $c$ )` over frozen ResNet-18 backbone features. The head is warm-started with cross-entropy for 5 epochs on the union of all source data; gradients are then recomputed against a differentiable soft-macro-F1 loss [12] so that the direction matches the macro-F1 evaluation metric. Gradient dimension: $d_g = d_{\text{feat}} \cdot c + c$ (full weight and bias gradient).

Text classification (Amazon). A two-layer MLP head `LayerNorm  $\rightarrow$  Linear( $d_{\text{feat}}$ ,  $h$ )  $\rightarrow$  ReLU  $\rightarrow$  Dropout  $\rightarrow$  Linear( $h$ , 1)`. The head is warm-started in two phases: first with binary cross-entropy on the union of all source data, then fine-tuned with differentiable soft-F1 loss [12]. We store only the final-layer weight gradient ($d_g = h$) to keep per-query cost low. Target gradients are computed on stratified positive/negative samples and averaged across 5 random seeds to reduce variance [43].